

Theorem 1 (Darboux's Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Assume that the one-sided derivatives $f'(a)$ and $f'(b)$ exist.*

If k is any real number strictly between $f'(a)$ and $f'(b)$, then there exists a point $c \in (a, b)$ such that

$$f'(c) = k.$$

Proof. Without loss of generality, assume that

$$f'(a) < f'(b).$$

Let k be a real number satisfying

$$f'(a) < k < f'(b). \tag{1}$$

Our objective is to prove that there exists a point $c \in (a, b)$ such that

$$f'(c) = k.$$

Step 1: Define an Auxiliary Function

Define a function $g : [a, b] \rightarrow \mathbb{R}$ by

$$g(x) = f(x) - kx.$$

Since f is continuous on $[a, b]$ and differentiable on (a, b) , the same is true for g .

Differentiating,

$$g'(x) = f'(x) - k.$$

Step 2: Analyze the Endpoints

Evaluating at the endpoints:

1. At $x = a$,

$$g'(a) = f'(a) - k.$$

Using (1),

$$f'(a) < k,$$

hence

$$g'(a) < 0.$$

2. At $x = b$,

$$g'(b) = f'(b) - k.$$

Again using (1),

$$k < f'(b),$$

hence

$$g'(b) > 0.$$

Step 3: Apply the Extreme Value Theorem

Since g is continuous on the closed interval $[a, b]$, the Extreme Value Theorem guarantees that g attains an absolute minimum value on $[a, b]$.

Thus, there exists a point $c \in [a, b]$ such that

$$g(c) \leq g(x) \quad \text{for all } x \in [a, b].$$

Step 4: Show that the Minimum Cannot Occur at the End-points

We now prove that the minimum point c cannot be equal to a or b .

- **The minimum cannot occur at a .**

Since

$$g'(a) = \lim_{x \rightarrow a^+} \frac{g(x) - g(a)}{x - a} < 0,$$

there exists $\delta > 0$ such that whenever

$$a < x < a + \delta,$$

we have

$$\frac{g(x) - g(a)}{x - a} < 0.$$

Because $x - a > 0$, it follows that

$$g(x) - g(a) < 0,$$

and therefore

$$g(x) < g(a).$$

Hence $g(a)$ cannot be the absolute minimum value of g on $[a, b]$.

- **The minimum cannot occur at b .**

Similarly,

$$g'(b) = \lim_{x \rightarrow b^-} \frac{g(x) - g(b)}{x - b} > 0.$$

Therefore, there exists $\delta > 0$ such that whenever

$$b - \delta < x < b,$$

we have

$$\frac{g(x) - g(b)}{x - b} > 0.$$

Since $x - b < 0$, multiplying both sides by the negative quantity $(x - b)$ reverses the inequality:

$$g(x) - g(b) < 0.$$

Thus,

$$g(x) < g(b),$$

showing that $g(b)$ also cannot be the absolute minimum.

Since the minimum cannot occur at either endpoint, it must occur at some interior point

$$c \in (a, b).$$

Step 5: Apply Fermat's Theorem

Because g attains a local minimum at the interior point c and g is differentiable at c , Fermat's Theorem implies that

$$g'(c) = 0.$$

Substituting the definition of $g'(x)$,

$$f'(c) - k = 0.$$

Hence,

$$f'(c) = k.$$

This completes the proof.

□